

Term Test II

Date: 2082/11/08

Level BE

Programme BEIT

Semester VII

Full Marks

50

Time

1.5 hrs

Subject: Engineering Economics

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt Any Five questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Calculate nominal interest when effective interest is 7% that compounds monthly. You deposited Rs 4,000 monthly at bank account for 6% interest that compounds quarterly. Find future sum at the end of 5th year. [10]

- ✓ 2. Your college is considering to purchase a machine for Rs. 7,00,000 expecting salvage value Rs. 1,00,000 at the end of 5th year. The use of machine generates Rs. 2,50,000 net annual income. Calculate IRR and make also unrecovered balance graph. [10]

3. Find both types of B/C ratio and justify investment with the given cash flow information. [10]

Initial investment	Rs. 6, 00,000
Annual revenue	Rs. 1, 80,000
Annual cost	Rs. 60,000
Salvage value	Rs.1, 00,000
Useful life year	5
MARR	8%

4. Perform sensitivity analysis of the following project over the range of $\pm 30\%$ in parameters; [10]

i) Initial investment ii) Annual revenue iii) MARR. Determine more and less sensitive parameter.

Initial investment	= 11,500
Annual revenue	= 3,000
Salvage value	= 1,000
Useful life	= 6 years
MARR	= 10%

5. Determine economic service life (ESL) of the following asset. [10]

Initial cost = Rs. 25,000

Operating cost = Rs. 10,000 for the first year that increases by 15% thereafter.

Salvage value = Rs. 10,000 for the first year that decreases by Rs. 1,000 thereafter.

Useful life = 6 years

MARR = 12%

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- ✓ 6. An equipment is purchased for Rs. 10, 00,000. It is expected to generate income of Rs. 3, 00,000 per year during 5 years and corporate income tax rate is 25% per year. Under the study period depreciation follows MACRS method. Calculate ATCF. [10]

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7. Write short notes on ANY TWO [10]

- a- Manufacturing and Non-manufacturing cost
- b- Principles of engineering economics
- c- Tax

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1. Distinguish between nominal and effective interest rate with suitable example. You deposited Rs 1,00,000 at present for 5% interest that compounds quarterly. How much could you draw equally at the end of each for 5 years with that amount? [10]
2. Cosmos college is considering to purchase a machine for Rs. 6,00,000 expecting salvage value Rs. 1,00,000 at the end of 5th year. The use of machine generates Rs. 2,80,000 annual income form Rs. 50,000 annual operating cost. Calculate IRR and B/C ratio for investment decision when MARR = 8%. [10]
3. Briefly explain the concept of breakeven analysis. A company wants to produce child computers with increasing market demand. The company is estimated following figures, [10]
 - Total cost = Rs. 25, 00,000
 - Fixed cost = Rs. 20,00,000
 - Total sales = Rs. 30, 00,000
 - Total quantity of output = 2500 units
 - a) Find break-even unit of output.
 - b) What should be the output if company desires profit of Rs. 10, 00,000?
4. Perform sensitivity analysis of the following project over the range of $\pm 30\%$ in parameters; i) Initial investment ii) Annual revenue iii) Life year. Determine more and less sensitive parameter. [10]
 - Initial investment = 15,000
 - Annual revenue = 4,000
 - Salvage value = 1,000
 - Useful life = 6 years
 - MARR = 10%
5. Determine economic service life (ESL) of the following asset. [10]
 - Initial cost = Rs. 10,000
 - Operating cost = Rs. 2,200 for the first year that increases by 20% thereafter.
 - Salvage value = Rs. 6,000 for the first year that decreases by 15% thereafter.
 - Useful life = 8 years
 - MARR = 15%
6. An equipment is purchased for Rs. 12, 00,000. It is expected to generate income of Rs. 4, 00,000 per year during 5 years and corporate income tax rate is 25% per year. Under the study period depreciation follows MACRS method. Calculate ATCF. [10]
7. Write short notes on **ANY TWO** [10]
 - a) Demand and supply
 - b) Principles of economics
 - c) Direct Tax and Indirect Tax